

Disproof of Conjecture 00000005344

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Counterexample

Every ordinary category can be regarded as a strict bicategory whose only 2-morphisms are identities. In this locally discrete bicategory, horizontal composition of 1-morphisms is exactly ordinary composition. Thus a category with noncommuting endomorphisms is enough to refute strict commutativity.

Use the category of sets and the Boolean set $B = \{\text{false}, \text{true}\}$. Define endomorphisms

$$f(x) = \text{false}, \quad g(x) = \neg x.$$

At $x = \text{false}$ the two composite orders give

$$(f \circ g)(x) = \text{false}, \quad (g \circ f)(x) = \text{true}.$$

Hence $f \circ g \neq g \circ f$.

Proposition 1. *Horizontal composition in a bicategory is not strictly commutative in general.*

Proof. Regard the category of sets as the locally discrete strict bicategory just described. The Boolean endomorphisms f and g have unequal horizontal composites, as witnessed at false. $\square$

The bicategory axioms provide associativity and units (up to coherent 2-isomorphism in the general case); they do not provide commutativity. Nor can normalizing a numerical constant to one turn two unequal 1-morphisms into equal ones.

The Lean project kernel-checks the two function evaluations, their inequality, and the negation of universal strict commutativity.